

Amrith Akshintala

Georgetown, TX | 512-888-4070 | aamrith@tamu.edu | [linkedin.com/in/aamrith](https://www.linkedin.com/in/aamrith) | github.com/CodingMastermind123 | [Portfolio](#)

TECHNICAL SKILLS

Programming: C++, Embedded C/C++, Python, Java, SQL, JavaScript, HTML/CSS, Swift
Embedded & Hardware: Arduino, Microcontrollers, Sensor Integration, PID Control, IMU, Circuit Design, 3D Printing
CAD: SolidWorks, AutoCAD
Robotics & Software: ROS2, Ubuntu/Linux, Git, React, Node.js

EDUCATION

Texas A&M University College Station, TX
Bachelor of Science in Computer Engineering with Honors Expected May 2029

- **Achievements:** Engineering Honors; Eagle Scout; AP Scholar with Distinction; National Merit Commended Scholar; PVSA (Presidential Volunteer Scholars Award) Gold
- **Relevant Coursework:** Multivariable Calculus, Electricity and Magnetism (Completed); Programming Design and Concepts, Discrete Structures, Digital Systems Design (in progress)
- **Organizations:** Texas A&M TURTLE Robotics (SNOUT R&D Team), Texas A&M SEDS (Lunabotics Team), Aggies Create (AirOne Team)

EXPERIENCE

Undergraduate Researcher Sep 2026 – Present
Lyu Lab, Texas A&M University (Dept. of Computer Science & Engineering) College Station, TX

- Conducting research on multi-robot coordination under Dr. Yiwei Lyu, focused on machine learning approaches to autonomous multi-agent systems

Undergraduate Researcher Aug 2025 – Apr 2026
Texas A&M Ultrasound Research Laboratory, under Dr. Sapna Bisht College Station, TX

- Developed non-invasive ultrasound-based measurement techniques to quantify tumor pressure
- Designed CAD models for treatment prototypes

PROJECTS

IMPACT-Robotics Project (ENGR 289) | *AutoCAD, Prototyping, C++, Arduino* Aug 2026 – Present

- Collaborating with a 3-person multidisciplinary team, mentored by TURTLE Robotics upperclassmen, to design and build an autonomous line-following robot for a competition course
- Designing an IR-sensor-based line-following system to navigate a branching track with turns and forks
- Engineering a multi-stage retrieval mechanism (scoop/claw and extending lift arm) to collect objects at three increasing platform heights, iterating through CAD modeling and 3D-printed prototyping

2-Axis Gimbal | *Arduino, C++, GY-87 IMU, SG90 Servos, SolidWorks* May 2026 – Present

- Designing a 2-axis camera gimbal with a team of 4; implemented closed-loop PID stabilization in C++ on an Arduino microcontroller using GY-87 IMU feedback to drive SG90 servos
- Completed full CAD modeling and assembly in SolidWorks; currently in final wiring and hardware assembly

Sensor Dashboard | *React, Node.js, Express, WebSocket, Recharts* Jul 2026

- Built a full-stack real-time telemetry dashboard for embedded sensor data, with a Node.js/serialport bridge streaming Arduino Uno R4 data through an Express + WebSocket backend to a React/Vite frontend
- Implemented live charting with Recharts, a drag-and-resize dashboard layout using react-grid-layout, and CSV data logging via a custom recorder module

AirOne (Aggies Create) | *Arduino, Embedded C++, Circuit Design* Sep 2026 – Present

- Working on the electrical subsystem of a combined laryngoscope/suction intubation medical device: Arduino-based sensing and control, battery system, user interface, and pump/motor integration

Lunabotics (TAMU SEDS, GNC Subteam) | *ROS2, Ubuntu, Python/C++* Sep 2026 – Present

- Evaluating candidate algorithms for the autonomous construction robot's GNC stack ahead of a team trade study: Kalman/EKF state estimation, A*/Dijkstra path planning, Pure Pursuit/Stanley path tracking, and PID control